Yoshwan Pathipati

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Education

Virginia Tech Blacksburg, VA
B.S. in Computer Science; Minors in Artificial Intelligence, Cybersecurity, and Mathematics *Aug 2023 – May 2027*
- **Honors:** Dean's List (top 20% of class) **Leadership:** Co-founder, Collaborative Coding Activities Club (80+ members)
- **Coursework:** Machine Learning, Intro to AI, Data Structures & Algorithms, Cloud Software Development, Cryptography

Technical Skills

Languages: Python, Java, C, JavaScript, TypeScript, SQL, Ruby, Bash, PowerShell, HTML/CSS
AI/ML: Large Language Models (LLMs), Ollama, Llama 3, Retrieval-Augmented Generation (RAG), JAX
Cloud & DevOps: AWS (EC2, S3, IAM, VPC, Security Hub, GuardDuty), Azure, GCP, Terraform, Docker, CI/CD, Linux
Security: Vulnerability Assessment, Nessus, OpenVAS, Wireshark, nmap, tfsec, NIST SP 800-53, NIST RMF, CMMC
Web & Databases: React, Node.js, Express, Flask, Django, Ruby on Rails, REST APIs, PostgreSQL, MySQL, MongoDB, Redis
Certifications: AWS Certified Solutions Architect – Associate (Jul 2026); CompTIA Security+ (in progress)

Experience

AI Research Assistant, LLM Automation Sep 2026 – Present
Virginia Tech, Dept. of Engineering Education *Blacksburg, VA*
- Develop Python preprocessing and faculty-reviewed prompts that generate CATME feedback via Ollama for **100+ students**
- Assess Llama, Qwen, and Kimi on VT compute, guiding faculty to models that meet classroom performance and cost goals
- Containerize the feedback pipeline with Docker alongside faculty, standardizing setup for a 100+ student cohort

Cybersecurity Intern May 2026 – Aug 2026
Triple Point Security *Blacksburg, VA*
- Ran authenticated and unauthenticated Nessus scans on 5 hosts, finding **20+ critical and 84 high-severity** issues
- Compared self-hosted Llama 3 with cloud LLMs for vulnerability-report generation, exposing severity-classification errors, missed CVEs, and RAG reliability limits that led to updated detection rules and fewer false positives
- Built an AWS–Azure site-to-site VPN, fixing an IKE policy mismatch; provisioned AWS, Azure, and GCP with Terraform
- Kept a stalled OpenVAS assessment on schedule by driving scans over GMP from the CLI in its Docker deployment
- Resolved a NIST/CMMC mismatch across 4 publications, setting the control structure for Security Hub compliance mapping
- Introduced EventBridge and SNS to the compliance team and trained 3 teammates to deploy workflows across AWS Regions

Research Assistant, Full-Stack Development (SpaceNet Testbed) Sep 2025 – Present
Hume Center, National Security Institute & Dept. of Aerospace and Ocean Engineering *Blacksburg, VA*
- Lead development of SpaceNet, a satellite-constellation simulation testbed, building its Flask REST APIs for 5 researchers
- Replaced SSH config editing with a browser workflow and 4 React telemetry dashboards used by 10+ team members
- Dockerized the platform and automated CI/CD with GitHub Actions, cutting deployment time from **45 to 8 minutes**
- Presented SpaceNet as a research poster at the Virginia Tech Summer Research Conference (2026)

Research Assistant, Backend Development (OpenDSA) Jun 2025 – Aug 2025
Virginia Tech, Dept. of Computer Science *Blacksburg, VA*
- Refactored 3 Rails controllers in OpenDSA's analytics modules, removing N+1 queries and cutting query time by **30%**
- Built Rails features for CS 3114's learning system: automated grading and real-time feedback that reduced manual grading

Python Developer Intern Jul 2024 – Oct 2024
Avenues, IIT Bombay *Remote*
- Shipped Django and MySQL backend features (authentication, profiles, data validation) across three core modules
- Found 6 critical auth-flow bugs through API testing with Postman and the DRF test client, all fixed within two sprints

Projects

Organic Farm E-Commerce Platform | *React, Node.js, Express, MongoDB, Stripe* Oct 2025 – Present
- Building a full-stack storefront for my family's farm business with listings, inventory management, and order processing
- Integrated Stripe payments and real-time inventory for **100+ produce SKUs**, replacing spreadsheet-based order-taking
- Mobile-first for the roughly 70% of customers who shop on phones, with a guest demo mode to try ordering without signing up

CloudGuard: Cloud Security Posture Monitoring | *Terraform, AWS, Security Hub, tfsec* Jan 2026 – Apr 2026
- Provisioned AWS (VPC, EC2, S3, IAM) with Terraform, wiring GuardDuty, Inspector, and Security Hub for threat detection
- Mapped every Security Hub finding to its NIST SP 800-53 control on a live React dashboard
- Added tfsec to a GitHub Actions pipeline that scans every push and blocks merges with infrastructure misconfigurations

Multimodal Associative Memory | *Python, JAX, Modern Hopfield Networks, Equilibrium Propagation* Feb 2026 – Present
- Designing an energy-based audiovisual associative memory that completes patterns from audio to vision and back
- Implemented continuous-time dynamics in JAX with a Modern Hopfield core using a Log-Sum-Exp energy function
- Trained with Equilibrium Propagation's local, biologically inspired weight updates instead of backpropagation